

CMPSC 465
Spring 2025
Final Exam

Name: _____

Time Limit: 110 Minutes

PSU Email ID: _____

This exam contains 14 pages (including this cover page, double-sided) and 4 questions.
Total of points is 52.

0. (0 points) (**Acknowledgement**) In this exam, you should write your solutions within the provided answer lines (for multiple choice questions) and solution boxes (for open-ended questions). Anything written outside these areas will not be graded no matter how correct they are.

(*Hint 1:* if you feel your solution is too long to fit in the provided solution box, then it is probably not be on the right track.)

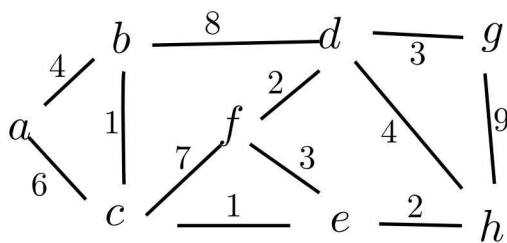
(*Hint 2:* if you feel your solution is too short for solution box, it's totally OK. Don't feel your solution is wrong because of its length. Some solutions can be described in two sentences, while I still give you a solution box of a full page.)

Write (copy) the following sentence in the following box. **Failed to do so will receive 0 for the final.**

“I acknowledge that I have to write my solutions within the provided answer lines and solution boxes. Anything written outside these dedicated areas will not be graded.”

1. (24 points) MCQ. There is exactly **one** correct choice for each question.

(1) (2 points) What is weight of a minimum spanning tree of the following graph?



- A. 15
- B. 17
- C. 14
- D. 16

(1) **D**

(2) (2 points) Consider a weighted and undirected graph with positive weights. If the weight of each edge is doubled, which of the following is true?

- A. The minimum spanning tree will change.
- B. The shortest path between any two nodes will change.
- C. Both the minimum spanning tree and shortest paths will remain the same.
- D. The minimum spanning tree will remain the same, but the shortest paths may change.

(2) **C**

(3) (2 points) Which of the following codes is possibly generated by Huffman's algorithm

- A. {0, 10, 11}
- B. {0, 1, 00, 111}
- C. {00, 01, 10, 110, 111}
- D. {1, 01, 001, 000}

(3) **A**

(4) (2 points) What is the minimum edit distance between "ALGORITHM" and "LOG-ARITHM"?

- A. 1
- B. 2
- C. 3
- D. 4

(4) _____ **C** _____

(5) (2 points) Which of the following statement about the 0-1 knapsack problem is true?

- A. The dynamic programming algorithm we learned in the lecture returns an optimal solution.
- B. The dynamic programming algorithm we learned in the lecture runs in polynomial time.
- C. The greedy algorithm we saw in the lecture returns an optimal solution.
- D. The greedy algorithm we saw in the lecture returns a solution with a bounded approximation ratio.

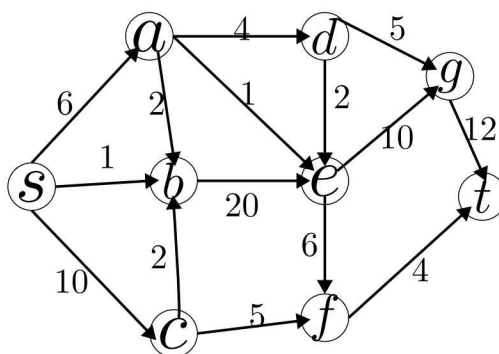
(5) _____ **A** _____

(6) (2 points) Which of the following statement about dynamic programming is true?

- A. Dynamic programming algorithms are always efficient (running in polynomial time).
- B. If a dynamic programming algorithm fills in an M -by- N table, then its running time is $O(MN)$.
- C. If a problem can be solved by dynamic programming, it can also be solved using divide and conquer.
- D. In dynamic programming, each problem is solved only once.

(6) _____ **D** _____

(7) (2 points) For the following flow network, with edge capacities as shown. What is the capacity of the minimum s - t cut?



- A. 11
- B. 12
- C. 13
- D. 14

(7) _____ **C** _____

- (8) (2 points) Given a flow network, if the capacity of every edge is decreased by 1 (but still positive), which of the following is true?

A. It is possible that the value of the maximum flow will remain the same.
B. The capacity of the minimum cut will be increased by at least 1.
C. The value of the maximum flow will be decreased by 1.
D. The value of the maximum flow will be decreased by at least 1.

(8) _____ **D** _____

- (9) (2 points) Suppose you want to maximize the function $f(x_1, x_2) = x_1 \cdot x_2 - \frac{1}{2}x_1 - \frac{1}{2}x_2$ on the regions $0 \leq x_1 \leq 10$ and $0 \leq x_2 \leq 15$. Which of the following is true?

A. The simplex algorithm can be used to solve this problem.
B. The objective function is nonlinear.
C. The feasible region is nonlinear.
D. This formulation is in the standard form of a linear programming formulation.

(9) _____ **B** _____

- (10) (2 points) Which of the following statement about linear programming is true?

A. If the feasible region is unbounded, then there is no solution.
B. Equality constraint is not allowed.
C. Any dual feasible solution is an upper bound on any primal feasible solution.
D. Any primal feasible solution is strictly larger than any dual solution.

(10) _____ **C** _____

- (11) (2 points) Let X and Y be two problems. If X is polynomial-time reducible to Y , which of the following is true?

A. If Y cannot be solved in polynomial time, then X cannot be solved in polynomial time.
B. X is harder than Y .
C. If X can be solved in polynomial time, then Y can be solved in polynomial time.
D. If Y can be solved in polynomial time, then X can be solved in polynomial time.

(11) _____ **D** _____

- (12) (2 points) Which of the following statements is false?

- A. It is known that **NP-complete** problems cannot be solved in polynomial time.
- B. The class **NP-complete** contains the hardest problems in **NP**
- C. All problems in **P** have an efficient certifier.
- D. All problems in **NP** have an efficient certifier.

(12) **A**

2. (24 points) A city plans to build libraries in n neighborhoods. Each neighborhood i has a population p_i . Due to noise restrictions, if a library is built in neighborhood i , no library can be built in neighborhoods $i - 1$ or $i + 1$. Design a dynamic programming algorithm to maximize the total population served by libraries.

To begin with, let us define the subproblem $L(j)$, as the maximum population served if libraries are placed in neighborhoods 1 to j .

- (1) (2 points) What is the final solution? Answer this question using the subproblem defined above.

Solution: $L(n)$

- (2) (5 points) What is the recurrence relation for $L(j)$?

Solution:

$$L(j) = \begin{cases} L(j-1) & \text{(no library in } j) \\ p_j + L(j-2) & \text{(library in } j) \end{cases}$$

- (3) (5 points) Specify the base case.

Solution: $L(0) = 0$ (no neighborhoods); $L(1) = p_1$ (only one neighborhood)

- (4) (5 points) Define an auxiliary table to reconstruct the optimal library placements.

Solution:

$$\text{prev}(j) = \begin{cases} j - 2, & \text{if } L(j) = p_i + L(j - 2) \\ j - 1, & \text{Otherwise} \end{cases}$$

- (5) (5 points) Explain how to recover the optimal locations using the auxiliary table defined above.

Solution: We first check if a library will be built on neighborhood n by checking if $L(n) = p_n + L(j - 2)$. Then we start with $\text{prev}(n)$ and do a back tracking (i.e., if we check $\text{prev}(j)$, we set $j' = \text{prev}(j)$ and next check $\text{prev}(j')$).

Note that it's also correct to define a True/False table in Part (4), but we still need to back track in this case.

- (6) (2 points) What is the running time of this dynamic programming algorithm? Briefly justify your solution.

Solution: $O(n)$, since there are $O(n)$ entries to compute and computing each entry takes $O(1)$ time.

3. (28 points) A university wants to assign n students to m dorm rooms, where each room can house at most one student. Each student has a list of preferred rooms. Assume $m \geq n$.
- (1) (8 points) Describe an algorithm that can quickly determine whether or not all students can be assigned to their preferred rooms. (*Hint: try to reduce this problem to the bipartite matching problem.*)

Solution: Construct a bipartite graph $G = (V, E)$, where V consists of the students and rooms. There's an edge between a student and a room when this room is in the preference list of the student. All students can be assigned to their preferred rooms if and only if there's a bipartite matching in G of size n .

- (2) (4 points) What is the running time of your algorithm? Justify your answer.

Solution: The bipartite graph G has $m + n$ vertices and $O(mn)$ edges. The bipartite matching algorithm in the lecture runs in time $O(|V||E|)$. So the running time is $O(mn(m + n))$.

- (3) (8 points) Now, let's take a digression to review a flow problem. Suppose you are given a directed graph $G = (V, E)$ with positive integer capacity c_e on each edge e , a source $s \in V$ and a sink $t \in V$. You are also given a maximum flow f in G . The flow is *acyclic*: there are no cycles in G on which all edges carry positive flow. The flow f is also integer valued.

In the homework, you showed that if the capacity of an edge $e^* \in E$ is reduced by 1, a maximum flow in the new flow network can be found in time $O(|E| + |V|)$.

Now, suppose the capacity of an edge e^* is increased by 1. Show that a maximum flow in the new flow network can still be found in time $O(|E| + |V|)$.

Solution: Assume the flow f is integer-valued. Let $e^* = (v, w)$. If $f(e^*) < c_{e^*}$, then increasing its capacity by 1 does not cause a problem.

Otherwise, if $f(e^*) = c_{e^*}$, increasing the capacity of e^* by 1 doesn't change the feasibility of the f , i.e., f is still a feasible flow. Then, we just build the residual graph G_f and try to find an augmenting path, and then update the flow if an augmenting path can be found.

Building G_f takes $O(|E| + |V|)$ time and find a path can be done using BFS. Hence, the total cost is $O(|E| + |V|)$.

- (4) (8 points) Back to the room assignment problem. Suppose a student changes their preferred room. Describe an efficient way to update the assignment without re-computing from scratch. Aim for an algorithm with running time $O(mn)$, which is faster than re-applying the algorithm in Part (1). (*Hint:* the results in Part (3) should help.)

Solution: Once a student changes his preferred room, we update the graph G built in Part (1), by removing and adding edges if necessary. Among the removed edges, at most one of them carried flow. And in the new graph, the size of the new matching cannot exceed the size of the old matching. So the size of the new matching is either unchanged, or reduced by 1. (Implied by Part (3)).

Then we just run one iteration of Ford-Fulkerson (or equivalently, updating the residual graph, finding an s - t path in the update residual graph), a new max flow can be found, which determined the size of the matching, which then further determines if all students can get their preferred room. Everything can be done in $O(|V| + |E|) = O(mn)$ time.